

SOUND - AN INTERACTIVE EBOOK



Kyle Forinash and Wolfgang Christian

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Introduction to Sound

Sound: An Interactive eBook

Sound: An Interactive eBook consists of 33 interactive simulations which require the reader to click buttons, move sliders, etc. in order to answer questions about the behavior of waves and sound in particular. There are also dozens of links to YouTube videos and other online resources that pertain to the topics being covered as well as suggestions for laboratory exercises and sound clips for understanding the fascinating subject of sound and music. The goal was to create an engaging text that integrates the strengths of printed, static textbooks and the interactive dynamics possible with simulations to engage the student in actively learning the physics of sound.

This book began in the way most textbooks do, as notes put together for a new course. The physics of sound, however, lends itself particularly well to examples, demonstrations and student participation in experiments. There are thousands of YouTube videos of interesting sound phenomena and dozens of simulations related to the physics of sound and music. This book was created from trying to provide access to these resources in a single source, first from a web page, then as interactive simulations on web pages and finally as this interactive textbook.

Interactivity

Gutenberg's invention of movable type around 1450 did not revolutionize the content or the format of the information being provided. It did have the important consequence of speeding up and broadening access to information. In a similar way, much of our modern technology has accelerated and expanded access to the world's knowledge base. Instructors today routinely provide a course syllabus, course information, instructor notes, assignments, sample tests, supplementary reading, and web links to other material, all online using a course management system or simple web pages. Many university students now receive access to a PDF version of the course textbook when they register for a course.

These uses, however, are *not* interactive. Much like an enhanced printing press, this technology serves to accelerate the one way transfer of material from the instructor to the student. In this regard it is not much different from what was already being done 560 years ago by Gutenberg; the information flow is unidirectional, albeit much faster. While, in hindsight, Gutenberg's creation was seminal to mass education, the communications revolution of the past century has yet to produce comparable improvements in human learning. However, today's technology has the capacity to function much more interactively of which this book is an example.

Note

Not all platforms allow JavaScript access to the sound hardware. Reset the simulation and try the Sound button again if the sound fails to play.

The above simulation (found in Chapter 10) is an example of the interactive nature of the simulations in this book. Play the simulation (you may have to click the reset button for the sound to turn on). What do you hear? What is the mathematical relationship between the beats you hear and the two frequencies? Use the slider to gradually change the difference in frequencies. At what point do you no longer hear beats? This unpleasant sound is called *dissonance*. At what difference in frequencies does this occur? If you continue to separate the two frequencies you will eventually hear two separate tones. When does this occur?

Acknowledgements

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TABLE OF CONTENTS

Introduction to Sound

Licensing

3.1: Basics

- 3.1.1: Basics- Work and Energy
 - 3.1.1.1: Motion and Force
 - 3.1.1.2: Newton's Three Laws
 - 3.1.1.3: Density and Pressure
 - 3.1.1.4: Energy and Power
 - 3.1.1.5: Two Energy Laws
 - 3.1.1.6: Conservation of Energy Simulation
- 3.1.2: Basics- The Molecular Basis of Matter
 - 3.1.2.1: Molecular Dynamics Simulation

3.2: Vibrations

- 3.2.1: Vibrations
 - 3.2.1.1: Simple Harmonic Motion
 - 3.2.1.2: Period, Frequency, Amplitude, Restoring Force, Phase
 - 3.2.1.3: Simple Harmonic Motion Simulation

3.3: Resonance

- 3.3.1: Resonance
 - 3.3.1.1: Resonance Examples
 - 3.3.1.2: A Few Other Examples of Resonance
 - 3.3.1.3: Harmonic Motion and Resonance Simulation
- 3.3.2: Resonance Springs
 - 3.3.2.1: Driven Springs Simulation
- 3.3.3: Quality Factor
 - 3.3.3.1: Quality
 - 3.3.3.2: Q-Factor Simulation

3.4: Wave Types

- 3.4.1: Transverse Waves
 - 3.4.1.1: Transverse Waves
 - 3.4.1.2: Transverse Wave Simulation
- 3.4.2: Longitudinal Waves
 - 3.4.2.1: Longitudinal Waves
 - 3.4.2.2: Longitudinal Wave Simulation
- 3.4.3: Other Waves
 - 3.4.3.1: Torsional Waves
 - 3.4.3.2: Examples of Waves
 - 3.4.3.3: Water Wave Simulation

- 3.4.4: Electromagnetic Waves
 - 3.4.4.1: Electromagnetic Waves
 - 3.4.4.2: Antenna Simulation

3.5: Wave Speed

- 3.5.1: Wave Speed
 - 3.5.1.1: Speeds of Different Types of Waves
 - 3.5.1.2: Speed of a Wave Simulation

3.6: Wave Behavior

- 3.6.1: Doppler Shift
 - 3.6.1.1: Doppler Shift
 - 3.6.1.2: The Doppler Effect Simulation
- 3.6.2: Mirrors and Reflection
 - 3.6.2.1: Reflection
 - 3.6.2.2: Reflection Simulation
- 3.6.3: Refraction
 - 3.6.3.1: Refraction
 - 3.6.3.2: Refraction Simulation
- 3.6.4: Lenses
 - 3.6.4.1: Lens Simulation
- 3.6.5: Dispersion
 - 3.6.5.1: Dispersion Simulation
- 3.6.6: Adding Wave Pulses
 - 3.6.6.1: Adding Waves
 - 3.6.6.2: Adding Two Wave Pulses (Superposition)
- 3.6.7: Adding Sinusoidal Waves
 - 3.6.7.1: Adding Two Linear Waves Simulation
- 3.6.8: Path Difference
 - 3.6.8.1: Interference
 - 3.6.8.2: Interference Due to Path Difference Simulation
- 3.6.9: Interference
 - 3.6.9.1: Interference- Ripple Tank Simulation
- 3.6.10: Diffraction
 - 3.6.10.1: Diffraction
 - 3.6.10.2: Diffraction and Diffraction Simulation

3.7: Pitch, Loudness and Timbre

- 3.7.1: Pitch, Loudness and Timbre
 - 3.7.1.1: Pitch
 - 3.7.1.2: Loudness
 - 3.7.1.3: The Decibel Scale
 - 3.7.1.4: Just Noticeable Difference
 - 3.7.1.5: Timbre (the first time)

3.8: Fourier Series

- 3.8.1: Sound Texture
 - 3.8.1.1: Wave Shape
 - 3.8.1.2: Sound Waveforms Simulation
- 3.8.2: Fourier Series
 - 3.8.2.1: Fourier Series
 - 3.8.2.2: Fourier Analysis
 - 3.8.2.3: Fourier Series Simulation
 - 3.8.2.4: Timbre (again!)
- 3.8.3: Microphone Sound Analyzer
 - 3.8.3.1: Fourier Analysis of Microphone Data

3.9: The Ear and Perception

- 3.9.1: The Ear and Perception
 - 3.9.1.1: Structure of the Ear
 - 3.9.1.2: The Place Theory of Hearing
 - 3.9.1.3: The Temporal Theory of Hearing
 - 3.9.1.4: Hearing Loss
 - 3.9.1.5: The Missing Fundamental
 - 3.9.1.6: Missing Fundamental Simulation
- 3.9.2: Beats
 - 3.9.2.1: Other Combination Tones
 - 3.9.2.2: Beats and Critical Bands Simulation
 - 3.9.2.3: Other Interesting Auditory Phenomena
 - 3.9.2.4: Animal Hearing

3.10: Strings

- 3.10.1: Driven String and Resonance
 - 3.10.1.1: String Resonance
 - 3.10.1.2: Driven String Simulation
- 3.10.2: Plucked String
 - 3.10.2.1: Plucked String
 - 3.10.2.2: Plucked String Simulation
- 3.10.3: Vibrating Plates Simulation
 - 3.10.3.1: Surface Resonances
 - 3.10.3.2: Vibrating Plate Simulation
 - 3.10.3.3: Other Surfaces
 - 3.10.3.4: Volume or Helmholtz Resonance
 - 3.10.3.5: Stringed Instruments

3.11: Tubes

- 3.11.1: Standing Waves in a Tube
 - 3.11.1.1: Tube Resonance
 - 3.11.1.2: Standing Wave Simulation
- 3.11.2: Reflection of Waves at a Boundary
 - 3.11.2.1: Reflection of Waves
 - 3.11.2.2: Reflection Simulation

- 3.11.3: Impedance
 - 3.11.3.1: Impedance
 - 3.11.3.2: Impedance Simulation
 - 3.11.3.3: Woodwind Instruments
 - 3.11.3.4: Brass Instruments
 - 3.11.3.5: Pipe Organs and Other Reed Instruments

3.12: Percussion

- 3.12.1: Percussion and Drumheads
 - 3.12.1.1: Vibrating Membranes
 - 3.12.1.2: Circular Drum Head Simulation
 - 3.12.1.3: Drums
 - 3.12.1.4: Harmonic Percussion Instruments
 - 3.12.1.5: Other Musical Instruments

3.13: Voice

- 3.13.1: The Human Voice
 - 3.13.1.1: The Vocal Tract
 - 3.13.1.2: Vocal Formants
 - 3.13.1.3: Phonemes
 - 3.13.1.4: Singing
 - 3.13.1.5: Animal Sounds

3.14: Musical Scales

- 3.14.1: Musical Scales
 - 3.14.1.1: The Pythagorean Scale
 - 3.14.1.2: Equal Temperament
 - 3.14.1.3: Temperament Simulation

3.15: Acoustics

- 3.15.1: Acoustics
 - 3.15.1.1: 16.1.1-Acoustic Qualities
 - 3.15.1.2: Reverberation
 - 3.15.1.3: Other Acoustical Effects
 - 3.15.1.4: Sound Reinforcement
 - 3.15.1.5: Spatial Perception
 - 3.15.1.6: Animal Acoustics
 - 3.15.1.7: Other Acoustical Applications

3.16: Electricity and Magnetism

- 3.16.1: EandM- Ohm's Law
 - 3.16.1.1: Ohm's Law
- 3.16.2: EandM- Currents and Magnetic Fields
 - 3.16.2.1: Currents Cause Magnetic Fields
 - 3.16.2.2: Magnetic Field Simulation
- 3.16.3: EandM- Electric and Magnetic Forces
 - 3.16.3.1: Currents in a Magnetic Field MAY Experience a Force
 - 3.16.3.2: Electric and Magnetic Forces Simulation

- [3.16.4: EandM- Faraday's Law](#)
 - [3.16.4.1: A CHANGING Magnetic Field Can Cause Current to Flow](#)
 - [3.16.4.2: Faraday's Law Simulation](#)

3.17: Electronics

- [3.17.1: Electronics](#)
 - [3.17.1.1: Microphones](#)
 - [3.17.1.2: Recording](#)
 - [3.17.1.3: Amplification](#)
 - [3.17.1.4: Speakers](#)
 - [3.17.1.5: Transmission](#)

[Index](#)

[Index](#)

[Glossary](#)

[Glossary](#)

[Detailed Licensing](#)

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A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).